

Uniwersytet Bielsko-Bialski

Wydział Budowy Maszyn i Informatyki

Digital Signal Processing

Exercise number:

1

Topic of the exercise:

Spectral analysis of deterministic signals: DFT and IDFT in matrix notation (variant 7)

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1. Aim of the exercise

The aim is to use the discrete Fourier transform pair in matrix notation: to build the index matrix K and the Fourier matrix W, to synthesise a discrete-time signal with the IDFT x = (1/N) W X for different values of N and to plot the result.

2. Theory

DFT: $X[\mu] = \Sigma\, x[k]\, e^{-j2\pi k\mu/N}$, IDFT: $x[k] = (1/N)\, \Sigma\, X[\mu]\, e^{+j2\pi k\mu/N}$, k, μ = 0 … N-1.

In matrix form: $x_\mu = W^* x_k$ and $x_k = (1/N)\, W\, x_\mu$, where W = exp(j 2π/N · K) (applied element-wise) is the N×N Fourier matrix and $K = [0\ 1\ \dots\ N\text{-}1]^T \cdot [0\ 1\ \dots\ N\text{-}1]$ contains all products kμ. The columns of W are the DFT eigensignals; W is symmetric and $W^{-1} = W^H/N = W^*/N$.

3. Task (variant 7)

Variant 7: $x_\mu = [6, 8, 2, 4, 3, 4, 5, 0, 0, 0]^T$, i.e. $N_0 = 10$ coefficients. The matrices are shown for $N = N_0 = 10$. The signal is synthesised for N = 10, 16 and 20; for $N > N_0$ the coefficient vector is padded with zeros at the end (X[μ] = 0 for μ ≥ 10).

4. Results

Index matrix K for N = 10:

k\μ	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	10	12	14	16	18
3	0	3	6	9	12	15	18	21	24	27
4	0	4	8	12	16	20	24	28	32	36
5	0	5	10	15	20	25	30	35	40	45
6	0	6	12	18	24	30	36	42	48	54
7	0	7	14	21	28	35	42	49	56	63
8	0	8	16	24	32	40	48	56	64	72
9	0	9	18	27	36	45	54	63	72	81

Fourier matrix W = exp(j2πK/10) for N = 10 (values rounded to two decimals; $W[k, \mu] = e^{j2\pi k\mu/10}$):

k\μ	0	1	2	3	4	5	6	7	8	9
0	1	1	1	1	1	1	1	1	1	1
1	1	0.81+0.59j	0.31+0.95j	-0.31+0.95j	-0.81+0.59j	-1	-0.81-0.59j	-0.31-0.95j	0.31-0.95j	0.81-0.59j
2	1	0.31+0.95j	-0.81+0.59j	-0.81-0.59j	0.31-0.95j	1	0.31+0.95j	-0.81+0.59j	-0.81-0.59j	0.31-0.95j
3	1	-0.31+0.95j	-0.81-0.59j	0.81-0.59j	0.31+0.95j	-1	0.31-0.95j	0.81+0.59j	-0.81+0.59j	-0.31-0.95j
4	1	-0.81+0.59j	0.31-0.95j	0.31+0.95j	-0.81-0.59j	1	-0.81+0.59j	0.31-0.95j	0.31+0.95j	-0.81-0.59j
5	1	-1	1	-1	1	-1	1	-1	1	-1
6	1	-0.81-0.59j	0.31+0.95j	0.31-0.95j	-0.81+0.59j	1	-0.81-0.59j	0.31+0.95j	0.31-0.95j	-0.81+0.59j
7	1	-0.31-0.95j	-0.81+0.59j	0.81+0.59j	0.31-0.95j	-1	0.31+0.95j	0.81-0.59j	-0.81-0.59j	-0.31+0.95j
8	1	0.31-0.95j	-0.81-0.59j	-0.81+0.59j	0.31+0.95j	1	0.31-0.95j	-0.81-0.59j	-0.81+0.59j	0.31+0.95j
9	1	0.81-0.59j	0.31-0.95j	-0.31-0.95j	-0.81-0.59j	-1	-0.81+0.59j	-0.31+0.95j	0.31+0.95j	0.81+0.59j

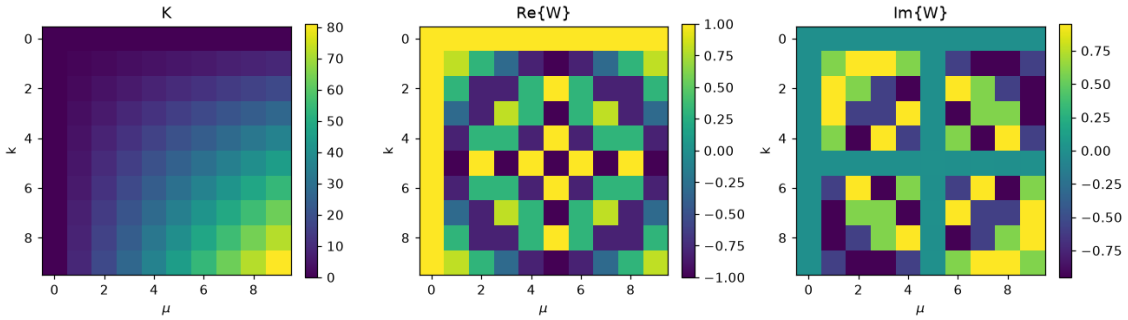


Figure 1. Index matrix K, real and imaginary part of the Fourier matrix W for N = 10.

Synthesised signal x[k] = (1/N) W X for N = 10:

k	0	1	2	3	4	5	6	7	8	9
Re x[k]	3.200	0.138	1.009	0.362	-0.109	0.000	-0.109	0.362	1.009	0.138
Im x[k]	0.000	0.923	0.833	0.218	0.778	-0.000	-0.778	-0.218	-0.833	-0.923
|x[k]|	3.200	0.934	1.309	0.422	0.786	0.000	0.786	0.422	1.309	0.934

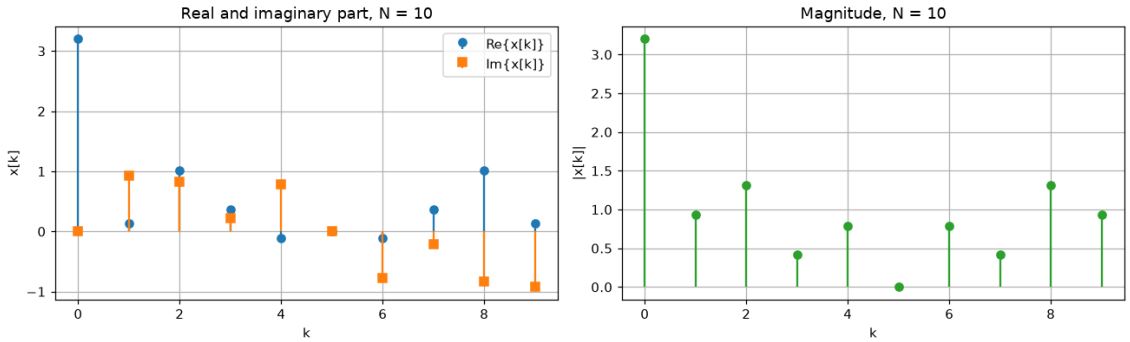


Figure 2. Synthesised signal for N = 10: real and imaginary part (left), magnitude (right).

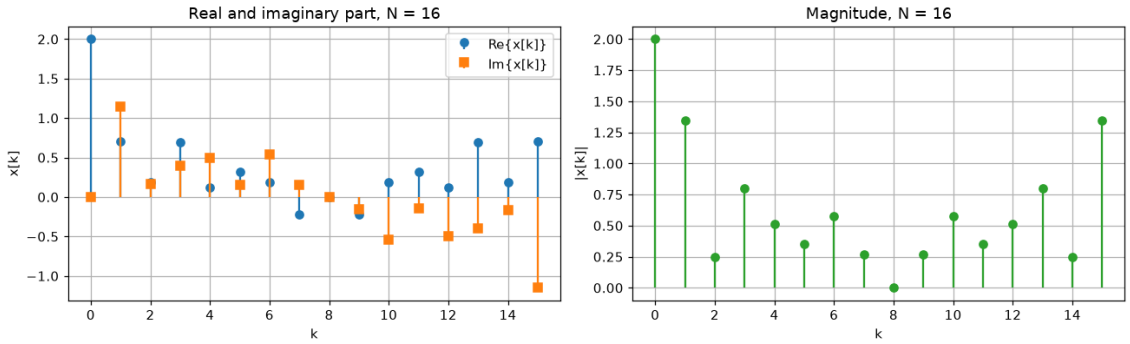


Figure 3. Synthesised signal for N = 16 (X zero-padded).

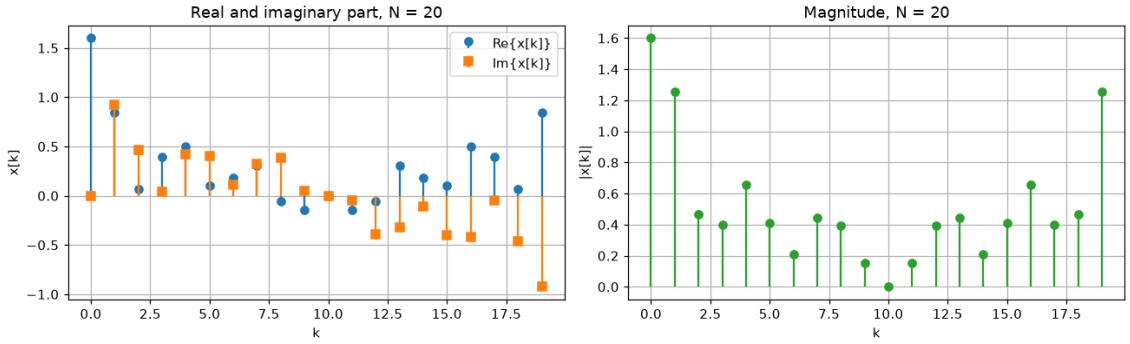


Figure 4. Synthesised signal for N = 20 (X zero-padded).

5. Interpretation

- Complex or real? The coefficients are real, but $X[N-\mu] \neq X[\mu]^*$ (e.g. X[1] = 8, X[9] = 0), so the synthesised signal is complex-valued. For N = 10 the signal has the symmetry $x[N-k] = x[k]^*$: the real part is even and the imaginary part is odd around k = 5 (e.g. x[1] = 0.138+0.923j, x[9] = 0.138-0.923j). The values x[0] = 32/10 = 3.2 and x[5] = (6-8+2-4+3-4+5)/10 = 0 are real.
- Check: for N = 10, 16 and 20 the result of the matrix multiplication agrees with numpy.fft.ifft to about 1e-15; $W^H W = N \cdot I$, W is symmetric and $W^{-1} = W^*/N$, as stated in the theory. The mean of x[k] equals X[0]/N = 0.6.
- Influence of N: X[μ] multiplies the eigensignal with μ periods per N samples, so zero-padding the coefficient vector to a larger N places the same coefficients at lower digital frequencies $2\pi\mu/N$ and the signal contains fewer periods of each component; its shape changes and it is longer. Because of the factor 1/N the amplitudes drop: x[0] = 32/N = 3.2 (N = 10), 2.0 (N = 16), 1.6 (N = 20). For N > 10 the vector X is no longer conjugate-symmetric in the DFT sense, so the signal does not have the symmetry of the N = 10 case and stays complex.

6. Conclusions

The IDFT in matrix notation reproduces the result of the FFT routine exactly (to rounding error). The signal for variant 7 is complex-valued because the coefficient vector is not conjugate-symmetric, and a change of N changes its length, amplitude scale (1/N) and the frequencies assigned to the coefficients.

Repository: https://github.com/RadoslawWalczak63/dsp_intro